sales

November 5, 2025

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import datetime as dt
```

[2]:
```
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
```

[3]:
```
df = pd.read_csv("sales_data_sample.csv", encoding='latin-1')
```

[4]:
```
print(df.head())
```

```
   ORDERNUMBER  QUANTITYORDERED  PRICEEACH  ORDERLINENUMBER    SALES  \
0        10107               30      95.70                2  2871.00
1        10121               34      81.35                5  2765.90
2        10134               41      94.74                2  3884.34
3        10145               45      83.26                6  3746.70
4        10159               49     100.00               14  5205.27

         ORDERDATE   STATUS  QTR_ID  MONTH_ID  YEAR_ID  …  \
0   2/24/2003 0:00  Shipped       1         2     2003  …
1    5/7/2003 0:00  Shipped       2         5     2003  …
2    7/1/2003 0:00  Shipped       3         7     2003  …
3   8/25/2003 0:00  Shipped       3         8     2003  …
4  10/10/2003 0:00  Shipped       4        10     2003  …

                    ADDRESSLINE1  ADDRESSLINE2           CITY STATE  \
0        897 Long Airport Avenue           NaN            NYC    NY
1             59 rue de l'Abbaye           NaN          Reims   NaN
2  27 rue du Colonel Pierre Avia           NaN          Paris   NaN
3            78934 Hillside Dr.            NaN       Pasadena    CA
4                7734 Strong St.           NaN  San Francisco    CA

  POSTALCODE COUNTRY TERRITORY CONTACTLASTNAME CONTACTFIRSTNAME DEALSIZE
0      10022     USA       NaN              Yu             Kwai    Small
1      51100  France      EMEA         Henriot             Paul    Small
2      75508  France      EMEA        Da Cunha           Daniel   Medium
3      90003     USA       NaN           Young            Julie   Medium
```

```
4        NaN      USA        NaN            Brown              Julie    Medium

[5 rows x 25 columns]
```

[5]:
```python
print(df.shape)
```

```
(2823, 25)
```

[6]:
```python
print(df.columns)
```

```
Index(['ORDERNUMBER', 'QUANTITYORDERED', 'PRICEEACH', 'ORDERLINENUMBER',
       'SALES', 'ORDERDATE', 'STATUS', 'QTR_ID', 'MONTH_ID', 'YEAR_ID',
       'PRODUCTLINE', 'MSRP', 'PRODUCTCODE', 'CUSTOMERNAME', 'PHONE',
       'ADDRESSLINE1', 'ADDRESSLINE2', 'CITY', 'STATE', 'POSTALCODE',
       'COUNTRY', 'TERRITORY', 'CONTACTLASTNAME', 'CONTACTFIRSTNAME',
       'DEALSIZE'],
      dtype='object')
```

[7]:
```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2823 entries, 0 to 2822
Data columns (total 25 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   ORDERNUMBER       2823 non-null   int64
 1   QUANTITYORDERED   2823 non-null   int64
 2   PRICEEACH         2823 non-null   float64
 3   ORDERLINENUMBER   2823 non-null   int64
 4   SALES             2823 non-null   float64
 5   ORDERDATE         2823 non-null   object
 6   STATUS            2823 non-null   object
 7   QTR_ID            2823 non-null   int64
 8   MONTH_ID          2823 non-null   int64
 9   YEAR_ID           2823 non-null   int64
 10  PRODUCTLINE       2823 non-null   object
 11  MSRP              2823 non-null   int64
 12  PRODUCTCODE       2823 non-null   object
 13  CUSTOMERNAME      2823 non-null   object
 14  PHONE             2823 non-null   object
 15  ADDRESSLINE1      2823 non-null   object
 16  ADDRESSLINE2      302 non-null    object
 17  CITY              2823 non-null   object
 18  STATE             1337 non-null   object
 19  POSTALCODE        2747 non-null   object
 20  COUNTRY           2823 non-null   object
 21  TERRITORY         1749 non-null   object
 22  CONTACTLASTNAME   2823 non-null   object
```

```
 23  CONTACTFIRSTNAME  2823 non-null   object
 24  DEALSIZE          2823 non-null   object
dtypes: float64(2), int64(7), object(16)
memory usage: 551.5+ KB
```

[8]:
```
# Select useful numeric columns
data = df[['QUANTITYORDERED', 'PRICEEACH', 'SALES']].copy()
```

[9]:
```
# Drop missing values if any
data.dropna(inplace=True)
```

[10]:
```
df.isnull().sum()
```

[10]:
```
ORDERNUMBER            0
QUANTITYORDERED        0
PRICEEACH              0
ORDERLINENUMBER        0
SALES                  0
ORDERDATE              0
STATUS                 0
QTR_ID                 0
MONTH_ID               0
YEAR_ID                0
PRODUCTLINE            0
MSRP                   0
PRODUCTCODE            0
CUSTOMERNAME           0
PHONE                  0
ADDRESSLINE1           0
ADDRESSLINE2        2521
CITY                   0
STATE               1486
POSTALCODE            76
COUNTRY                0
TERRITORY           1074
CONTACTLASTNAME        0
CONTACTFIRSTNAME       0
DEALSIZE               0
dtype: int64
```

[11]:
```
to_drop = ['ADDRESSLINE1', 'ADDRESSLINE2', 'STATE', 'POSTALCODE',
'PHONE']
df = df.drop(to_drop, axis=1)
```

[12]:
```
df.isnull().sum()
```

[12]:
```
ORDERNUMBER             0
QUANTITYORDERED         0
PRICEEACH               0
ORDERLINENUMBER         0
SALES                   0
ORDERDATE               0
STATUS                  0
QTR_ID                  0
MONTH_ID                0
YEAR_ID                 0
PRODUCTLINE             0
MSRP                    0
PRODUCTCODE             0
CUSTOMERNAME            0
CITY                    0
COUNTRY                 0
TERRITORY            1074
CONTACTLASTNAME         0
CONTACTFIRSTNAME        0
DEALSIZE                0
dtype: int64
```

[13]:
```
df.dtypes
```

[13]:
```
ORDERNUMBER           int64
QUANTITYORDERED       int64
PRICEEACH           float64
ORDERLINENUMBER       int64
SALES               float64
ORDERDATE            object
STATUS               object
QTR_ID                int64
MONTH_ID              int64
YEAR_ID               int64
PRODUCTLINE          object
MSRP                  int64
PRODUCTCODE          object
CUSTOMERNAME         object
CITY                 object
COUNTRY              object
TERRITORY            object
CONTACTLASTNAME      object
CONTACTFIRSTNAME     object
DEALSIZE             object
dtype: object
```

[14]:

```
scaler = StandardScaler()
scaled_data = scaler.fit_transform(data)
```

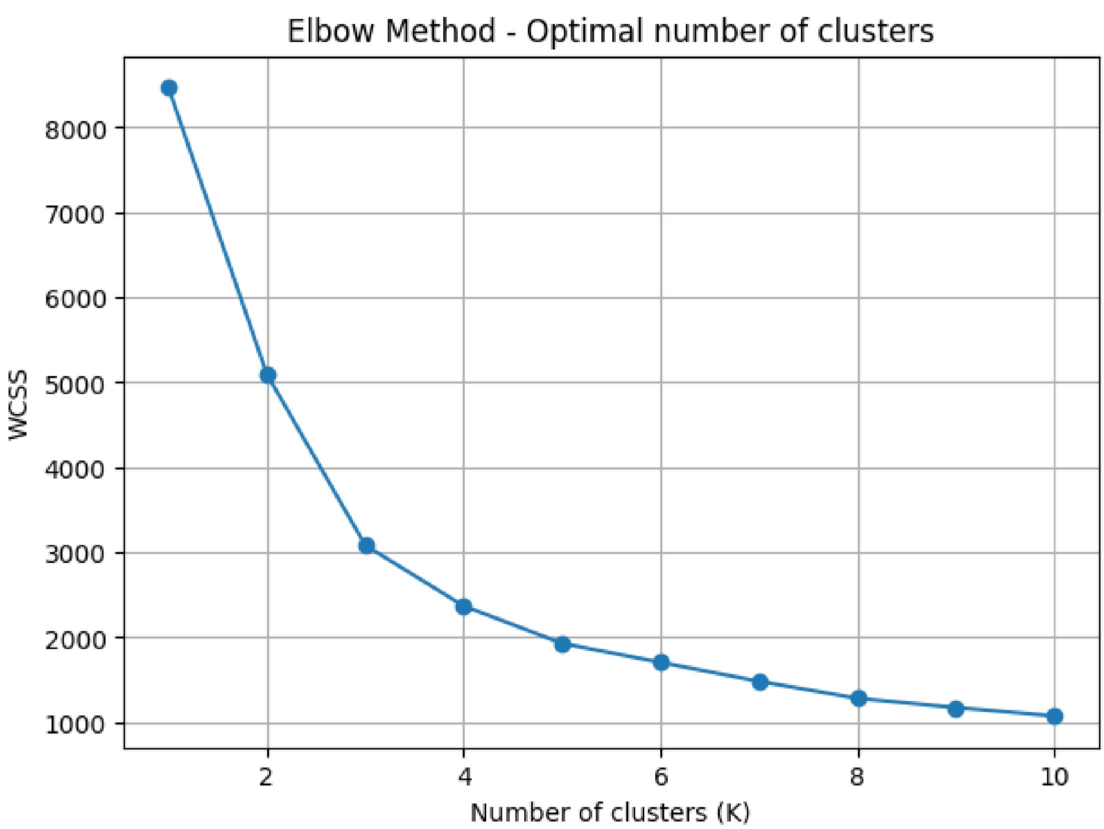
[15]:

```
wcss = []
K_range = range(1, 11)

for k in K_range:
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(scaled_data)
    wcss.append(kmeans.inertia_)
```

[16]:

```
plt.figure(figsize=(7,5))
plt.plot(K_range, wcss, marker='o')
plt.title("Elbow Method - Optimal number of clusters")
plt.xlabel("Number of clusters (K)")
plt.ylabel("WCSS")
plt.grid(True)
plt.show()
```



[17]:

```
optimal_k = 4
kmeans = KMeans(n_clusters=optimal_k, random_state=42)
cluster_labels = kmeans.fit_predict(scaled_data)

# Add cluster information to original dataframe
df['Cluster'] = cluster_labels
```

[18]:

```
df.head()
```

[18]:

```
   ORDERNUMBER  QUANTITYORDERED  PRICEEACH  ORDERLINENUMBER    SALES  \
0        10107               30      95.70                2  2871.00
1        10121               34      81.35                5  2765.90
2        10134               41      94.74                2  3884.34
3        10145               45      83.26                6  3746.70
4        10159               49     100.00               14  5205.27

         ORDERDATE   STATUS  QTR_ID  MONTH_ID  YEAR_ID  … MSRP  PRODUCTCODE  \
0   2/24/2003 0:00  Shipped       1         2     2003  …   95     S10_1678
1    5/7/2003 0:00  Shipped       2         5     2003  …   95     S10_1678
2    7/1/2003 0:00  Shipped       3         7     2003  …   95     S10_1678
3   8/25/2003 0:00  Shipped       3         8     2003  …   95     S10_1678
4  10/10/2003 0:00  Shipped       4        10     2003  …   95     S10_1678

              CUSTOMERNAME           CITY COUNTRY TERRITORY CONTACTLASTNAME  \
0        Land of Toys Inc.            NYC     USA       NaN              Yu
1       Reims Collectables          Reims  France      EMEA         Henriot
2          Lyon Souveniers          Paris  France      EMEA        Da Cunha
3        Toys4GrownUps.com       Pasadena     USA       NaN           Young
4  Corporate Gift Ideas Co.  San Francisco     USA       NaN           Brown

  CONTACTFIRSTNAME DEALSIZE Cluster
0             Kwai    Small       3
1             Paul    Small       3
2           Daniel   Medium       1
3            Julie   Medium       0
4            Julie   Medium       1

[5 rows x 21 columns]
```

[22]:

```
kmeans = KMeans(n_clusters=4, random_state=42)
df['Cluster'] = kmeans.fit_predict(scaled_data)
df.head()
```

[22]:

```
   ORDERNUMBER  QUANTITYORDERED  PRICEEACH  ORDERLINENUMBER    SALES  \
0        10107               30      95.70                2  2871.00
1        10121               34      81.35                5  2765.90
2        10134               41      94.74                2  3884.34
```

```
3        10145               45      83.26                6  3746.70
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         ORDERDATE   STATUS  QTR_ID  MONTH_ID  YEAR_ID  … MSRP  PRODUCTCODE  \
0   2/24/2003 0:00  Shipped       1         2     2003  …   95     S10_1678
1    5/7/2003 0:00  Shipped       2         5     2003  …   95     S10_1678
2    7/1/2003 0:00  Shipped       3         7     2003  …   95     S10_1678
3   8/25/2003 0:00  Shipped       3         8     2003  …   95     S10_1678
4  10/10/2003 0:00  Shipped       4        10     2003  …   95     S10_1678

              CUSTOMERNAME           CITY COUNTRY TERRITORY CONTACTLASTNAME  \
0        Land of Toys Inc.            NYC     USA       NaN              Yu
1       Reims Collectables          Reims  France      EMEA         Henriot
2          Lyon Souveniers          Paris  France      EMEA        Da Cunha
3        Toys4GrownUps.com       Pasadena     USA       NaN           Young
4  Corporate Gift Ideas Co.  San Francisco     USA       NaN           Brown

  CONTACTFIRSTNAME DEALSIZE Cluster
0             Kwai    Small       3
1             Paul    Small       3
2           Daniel   Medium       1
3            Julie   Medium       0
4            Julie   Medium       1

[5 rows x 21 columns]
```

[ ]: